



[HTTPS://MB-PORTFOLIO-ONE.VERCEL.APP/](https://mb-portfolio-one.vercel.app/)

CONTACT

+381 637220573

markobjelica4@gmail.com

Alekse Santica 68
21000 Novi Sad

GitHub:

<https://github.com/MarkoBjelica98>

Linkedin

www.linkedin.com/in/marko-bjelica-55143ab8

LANGUAGES

- English (C2)
- Serbian (Native)

SOFT SKILLS:

- Problem-solving mindset
- Fast learner
- Proactive
- Team collaboration
- Attention to detail

EDUCATION

- Faculty of Technical Sciences, Novi Sad
- Measurements and Regulation (Expected 2026)

MARKO BJELICA

SOFTWARE DEVELOPER

PROFILE

Full-stack developer with hands-on experience building admin dashboards, backend APIs, and real-time monitoring systems.

Worked on a machine learning admin panel that includes analytics, A/B testing, user management, and role-based access control.

Strong foundation in React, FastAPI, PostgreSQL, and embedded systems, with practical experience solving real-world issues such as API optimization, authentication, state management, and system integration.

Focused on growing into a backend/full-stack engineer role while working on scalable and data-driven systems.

WORK EXPERIENCE

● ML Monitoring Admin Panel (Full-Stack Internship)

- Built a full-stack admin dashboard for monitoring machine learning models and managing platform administration
- Developed backend APIs using FastAPI + SQLAlchemy and integrated them with a React frontend
- Implemented analytics modules including:
 - prediction trends over time
 - probability distribution
 - model performance metrics (ROC, Precision-Recall, confusion matrix)
- Designed and implemented a multi-level user and role management system for the ML platform, including role-based access control (RBAC) scoped at both organization (company) and project level, with roles such as Admin (full access to configuration, users, and models) and User (restricted access to model insights and analytics), enabling assignment, updating, and removal of roles, and enforcing permissions across the system
- Implemented authorization flows and permission checks across backend and frontend to ensure secure access based on user role and project context
- Built and integrated A/B testing configuration and statistics tracking
- Designed modular UI architecture with separate panels for:
 - metadata
 - trends
 - probability distribution
 - evaluation metrics
 - A/B testing
- Optimized frontend performance by reducing duplicate API calls and introducing panel-level loading instead of global loaders
- Identified and resolved issues across the system, improving stability, performance, and overall user experience

● IoT System Prototype

- Connected sensors to microcontroller and transmitted data to web dashboard
- Displayed real-time values using custom frontend
- Integrated embedded system with web backend
- Applied full data flow: device → backend → frontend

● Embedded Systems – EasyPIC PRO v7 + Compass 2

- Developed firmware in C for I2C communication
- Integrated compass module and processed real-time orientation data
- Implemented sensor data reading and debugging
- Worked with low-level microcontroller systems

● Personal Portfolio Website

- Built responsive portfolio using React (Vite) + Tailwind CSS
- Implemented reusable components and modern UI design
- Added animations, scroll effects, and optimized layout
- Used Git for version control

Frontend:

React (Vite), JavaScript (ES6+), Tailwind CSS, Recharts

Backend:

Python (FastAPI), SQLAlchemy, REST APIs

Database:

PostgreSQL, SQL (queries, indexing basics)

Embedded:

C / C++, mikroC PRO, I2C communication, PIC microcontrollers